

infl_ens: Influencer-Game Routing for Specialized SLMs

Project summary

September 14, 2026

Abstract

infl_ens asks whether small language-model adapters can specialize through a mathematically structured competition for data. It transfers the influencer’s game—a spatial game in which agents compete for a resource distribution—to alignment: resources are prompts, agents are LoRA adapters, and their positions lie in a learned safety trait space. A router allocates prompts, SFT changes the adapters, and positions are re-estimated. This is a controlled research hypothesis about specialization, not yet a finished alignment solution. Current runs show clear *routing* specialization, but mixed end-to-end model-quality results; that gap is the central question.

1 The project in one minute

Clone a base SLM several times and place the clones in a low-dimensional map of prompts. Prompts are resources. A clone near a prompt receives more of that training data, but competes with nearby clones. After training the clones on allocated data, refresh their positions and repeat:

corpus $\rightarrow$ trait positions $\rightarrow$ routing $\rightarrow$ LoRA SFT $\rightarrow$ updated positions.

The desired outcome is a routed ensemble of complementary specialists rather than nearly identical fine-tunes trained on one pooled mixture.

For mechanistic interpretability, the trait coordinates are not claimed to be literal features inside a transformer. They are an *external causal state abstraction*: a benchmark-supervised summary used to intervene on each adapter’s training distribution. The project tests whether that intervention produces detectable specialization in routing, cross-perplexity, and held-out benchmark NLL.

2 The source theory

The supplied paper, *Learning Dynamics of the Influencer’s Game in Resource Landscapes*, studies N agents choosing positions x_i . Resources have traits b and distribution $B(b)$. Agent i has influence $f_i(x_i, b)$ and receives a proportional share:

$$G_i(\mathbf{x}, b) = \frac{f_i(x_i, b)}{\sum_{j=1}^N f_j(x_j, b)}, \quad u_i(\mathbf{x}) = \sum_b B(b) G_i(\mathbf{x}, b). \quad (1)$$

The game is constant-sum. The historical implementation uses a Gaussian influence kernel,

$$f_i(x_i, b) = \exp\left[-\frac{1}{2}(x_i - b)^\top \Sigma^{-1}(x_i - b)\right], \quad (2)$$

with $\Sigma = \sigma^2 I$ when isotropic. Large σ means broad competitive reach (generalism); small σ means a narrow niche (specialism).

The kernel-agnostic implementation additionally supplies a smoothed multivariate hyperbolic kernel, a mode-parameterized Dirichlet kernel, and independent mode-parameterized beta factors. All are log-concave and expose $\nabla_x \log f$ and $\nabla_x^2 \log f$ to the same game dynamics. Dirichlet routing uses the native probability simplex; the kernel comparison maps every treatment to that same seven-dimensional domain rather than comparing different geometries.

Under suitable log-concavity conditions, the paper establishes a unique symmetric Nash equilibrium. With Gaussian influence, it is the resource-weighted mean. In one dimension the first stability boundary is

$$\sigma_0^* = \sqrt{\frac{N-2}{N-1}} \sigma_B, \quad (3)$$

where σ_B is resource spread. Above this threshold, gradient dynamics return agents to a shared generalist position. Below it, the symmetric state can lose stability and form specialist clusters, cycles, and smaller competitive subgames. The paper also reports the same qualitative bifurcation picture under independent Q-learning.

The alignment transfer is simple: a multi-modal prompt landscape should allocate different regions to different adapters when the router is narrow enough. The caveat is important: SFT does not literally execute the game’s spatial gradient flow. The code’s theory-versus-SFT experiments exist to test this proposed bridge.

3 What the repository implements

3.1 A learned safety trait space

The `data` package turns labeled benchmark families into a common coordinate system. Present axes cover harm (BeaverTails), hallucination (HaluEval), jailbreak behavior (JBB-Behaviors or ToxicChat), privacy (AI4Privacy), over-refusal (OR-Bench), prompt injection, and policy violation (Do-Not-Answer). Text is embedded with a sentence-transformer or Hugging Face encoder; learned label-derived directions project embeddings into $[0, 1]^L$. A configured softmax coordinate map instead produces a native probability-simplex representation. Corpus density becomes the empirical resource distribution.

3.2 Game-theoretic routing and LoRA training

The `inflgame` package implements Equation 1: configurable influence kernels, allocation G , expected utility, exact score-function utility gradients, and kernel-matched first-bifurcation thresholds. The hyperbolic threshold uses its generalized-eigenvalue closed form; Dirichlet and product-beta thresholds use numerical equilibrium and root solvers. Theory-only runs optimize positions by gradient ascent. Closed-loop runs dispatch real prompts to actual adapters.

Proportional routing assigns a prompt in proportion to G_i . *Strategic routing* uses $G_i(1 - G_i)$, the factor in the strategic gradient: it de-emphasizes examples an agent already dominates and emphasizes contested examples. The training package then runs SFT for each router agent, saves LoRA adapters under `results/<run>/agents/`, and refreshes positions from held-out projected prompts. It supports both cumulative and fresh-per-round LoRA, pooled-replay baselines, and pair-merge experiments that reduce two trained clones to one served expert.

Offline theory initialization and stability matching use the full calibrated KDE resource measure. The online game-gradient update intentionally does not assume that global knowledge: it estimates the same kernel-specific game gradient from the current observed training batch with uniform

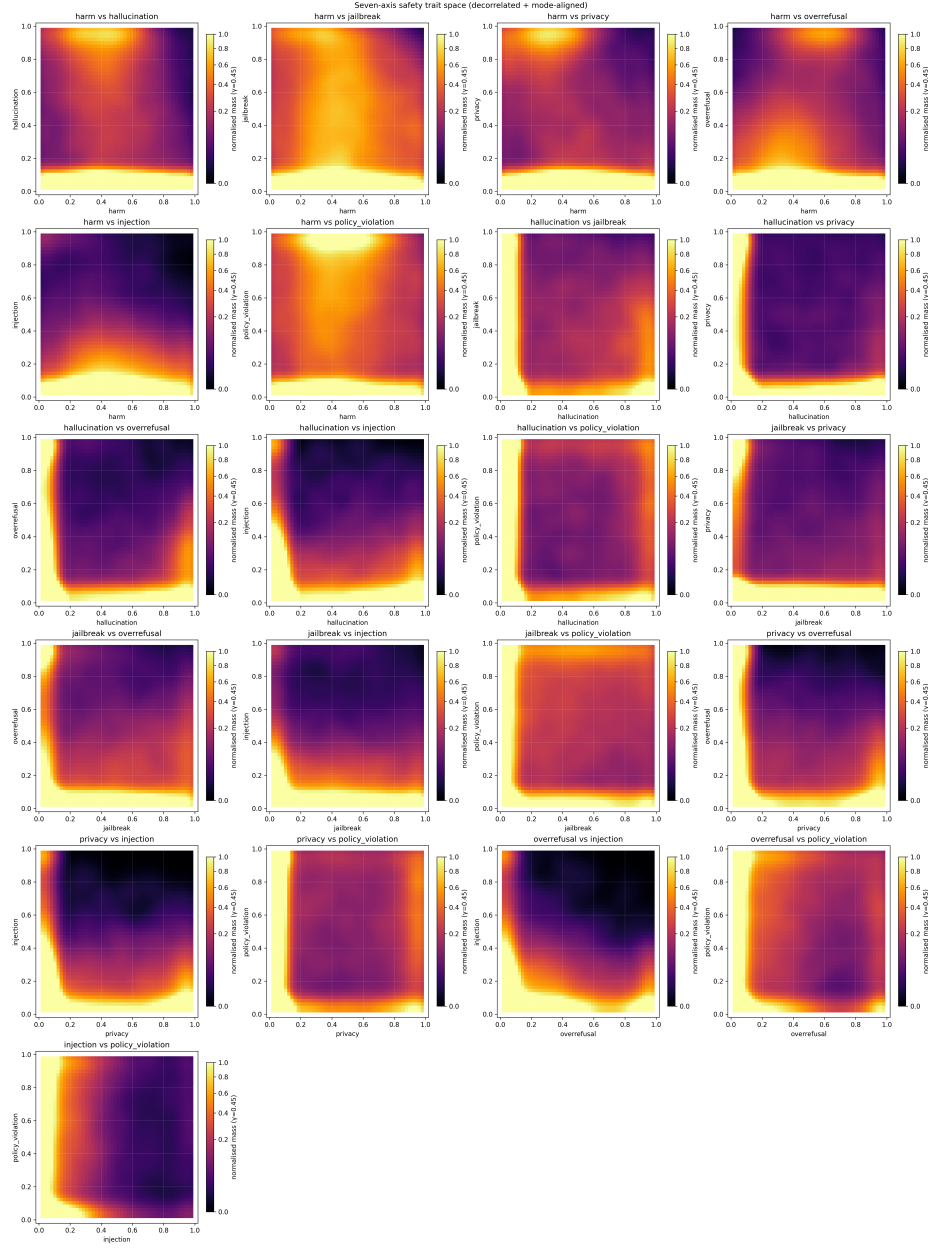


Figure 1: Pairwise slices of a seven-axis safety resource landscape generated by this repository. Bright regions are prompt-dense locations in an external learned trait map, not claimed model-internal features. This multi-modal geometry makes routing inspectable and supplies the resource landscape for the game.

empirical mass. An offline-KDE update remains available only as an oracle diagnostic. Soft top- k gates SFT assignment, not the position gradient.

Three objects must remain distinct: **trait position** (the router’s geometric summary), **adapter parameters** (the actual transformer changes), and **evaluation behavior** (NLL and specialist probes). Theory directly describes the first; the scientific question is whether it causally organizes the second enough to improve the third.

4 How to read the current evidence

The repo contains synthetic/theory-only routers, two- and three-axis safety spaces, four- and six-agent LoRA loops, fixed-theory AI4Privacy studies, and seven-axis pair-merge experiments. Evaluation reports mean next-token negative log likelihood (NLL) on chat-formatted benchmark splits. Lower NLL means higher probability for a reference continuation; it is useful for specialization diagnostics but not a complete safety or helpfulness metric.

The strongest positive result is geometric and attributional. In one recorded seven-axis run, the winning route matched benchmark type: the privacy expert won 858/1,000 AI4Privacy examples, the hallucination expert 825 HaluEval examples, the harm expert 623 BeaverTails examples, the policy expert 699 Do-Not-Answer examples, and the over-refusal expert 586 OR-Bench examples. This supports meaningful data specialization rather than arbitrary partitioning.

The end-to-end result is deliberately more cautious. In that run’s flat routing diagnostic, the pooled baseline had NLL 1.947, learned expected routing had 1.989, and an oracle choice among experts had 1.916. The ensemble contains potentially useful complementary experts, but the geometric router has not yet consistently converted that complementarity into a win over pooled training. Candidate causes include an insufficient trait map, a mismatch between NLL-optimal and influence-optimal routing, pair-merge dilution, or SFT dynamics that differ from the assumed position dynamics.

5 Why it is interesting for mechanistic interpretability

- **Controlled curriculum interventions:** changing σ , initialization, or routing policy changes the data distribution while holding base model and SFT fixed.
- **Predicted phase changes:** Equation 3 motivates a collapse-to-specialize transition rather than merely asserting that diversity helps.
- **Several observable levels:** geometry (positions/allocation), behavior (NLL/benchmark scores), and parameters or representations (LoRA deltas/activations).
- **Useful counterfactuals:** pooled replay, fixed-theory positions, random versus theory initialization, proportional versus strategic routing, and corner ablations.

An informative next study would match examples per adapter while changing only their trait-space curriculum. It could compare LoRA-delta similarity, layerwise activation differences on each benchmark, and whether a linear probe for an external trait coordinate becomes more separable in the expected specialist. A compelling result would connect a known curriculum intervention, a behavioral niche, and a localized parameter or representational change.

6 Caveats and next steps

1. **The theorem is not about transformers.** It proves properties of a spatial game; SFT is a proposed realization that needs empirical fidelity checks.
2. **External axes are imperfect proxies.** Labels and embeddings can confound topic, style, difficulty, and response format, so axis-separability diagnostics matter.
3. **NLL is limited.** It is not refusal quality, robustness, calibration, or adversarial safety; behavioral metrics are needed before strong alignment claims.
4. **Routing is the bottleneck.** The oracle-versus-router gap motivates a learned or calibrated router while retaining the game as a prior or regularizer.
5. **Attribution is decisive.** Use existing 2-by-2 controls, matched compute, multiple seeds, and held-out splits to identify what actually causes a gain.

7 Bottom line

`infl_ens` turns a theory of competitive niche formation into an experimental design for SLM specialization. Its router already finds semantically sensible niches, but those niches do not automatically outperform a well-trained pooled model. Explaining or closing that gap—with attributional controls and parameter/representation-level measurements—is the central next experiment.

Source. Based on the supplied Lovett and Fu draft, *Learning Dynamics of the Influencer’s Game in Resource Landscapes*, and the repository state inspected on September 14, 2026. [Supplied paper PDF](#).